Scenario: *modelling earthquake catastrophe risk across the Taupo Volcanic Zone (TVZ), New Zealand, using the 2007-2023 GeoNet earthquake catalogue to calibrate the model's seismicity assumptions. Portfolio assumption: NZD 1.2bn Total Insured Value across ~3,000 residential, commercial and geothermal-industrial risks.*

Exposure data and pre-modelling

- **COPE:** Construction, Occupancy, Protection, Exposure; the four core risk characteristics needed before a risk can be modelled.
  - ***Example:*** *a geothermal processing plant near Wairakei is coded as Reinforced Concrete construction (C), Industrial occupancy (O), sprinklered with a monitored alarm (P), and a NZD 42m Total Insured Value (E).*
- **TIV (Total Insured Value):** The full insured/replacement value of a risk; the base unit loss is measured against.
  - ***Example:*** *the TVZ portfolio totals NZD 1.2bn in TIV across ~3,000 risks spread between Taupo, Rotorua, Tokoroa and Kawerau.*
- **Slip:** The broker's placing document setting out policy terms, coverage and exposure information for a risk.
  - ***Example:*** *a broker's slip lands for the new NZD 85m Wairakei geothermal plant, requesting earthquake and business-interruption cover from January 1st.*
- **Geocoding / CRESTA zone:** Converting a risk's address into a precise location; CRESTA zones are the industry-standard geographic zones used to aggregate catastrophe exposure.
  - ***Example:*** *a residential risk in central Taupo is geocoded from its street address down to postcode, then assigned to an illustrative CRESTA-style zone covering the Central North Island, the resolution the Hazard module needs to apply the right ground-shaking intensity.*

Inside the model

- **Stochastic event catalogue:** A simulated set of thousands of possible catastrophe events used to represent the full range of plausible outcomes.
  - ***Example:*** *17 years of GeoNet data (2007-2023) is too short to capture the full range of possible TVZ earthquakes, so Touchstone draws on a simulated catalogue of thousands of plausible events, calibrated against that same historical record, to represent centuries of possible seismicity in a single model run. My own simulation below uses 50,000 simulated years as a stand-in for this.*
- **Hazard / Vulnerability / Financial module:** The three core model stages: simulating the physical peril, translating hazard into physical damage, then converting damage into an insured loss via policy terms.
  - ***Example:*** *for a simulated M6.8 rupture on the Taupo Fault Belt: the Hazard module estimates ground-shaking intensity at each location; the Vulnerability module converts that into a damage ratio (a timber-frame house responds very differently to a reinforced-concrete geothermal plant); the Financial module then applies the policy's deductible and limit to turn that physical damage into an insured loss.*
- **Secondary uncertainty and demand surge:** The spread of outcomes around the mean loss estimate, and the temporary post-event inflation of repair costs, respectively.
  - ***Example:*** *two identical simulated M6.5 events might still produce a materially different modelled loss purely from secondary uncertainty; if a real event also triggered a spike in North Island builder day-rates afterwards, demand surge would push the eventual claims cost higher still.*
- **Own View of Risk / Our View of Risk:** A (re)insurer's internally adjusted view of catastrophe risk, layered on top of the raw vendor model output to reflect claims experience, expert judgement or alternative model views.
  - ***Example:*** *if claims history showed TVZ geothermal assets perform worse in earthquakes than Touchstone's default vulnerability curves assume, Our View of Risk would apply an uplift to those specific curves before the output is trusted for pricing.*

Loss output and metrics

AAL (Average Annual Loss)

The long-run expected loss per year for a risk or portfolio; the core metric for technical pricing.

- **Example:** across the 50,000 simulated years in my TVZ earthquake run, the average loss per year comes out at approximately **NZD 6.2m (about 0.5% of the portfolio's NZD 1.2bn TIV),** this is the single number an underwriter would use as the baseline technical price for earthquake cover on this book, before loadings for expenses, profit and reinsurance cost.

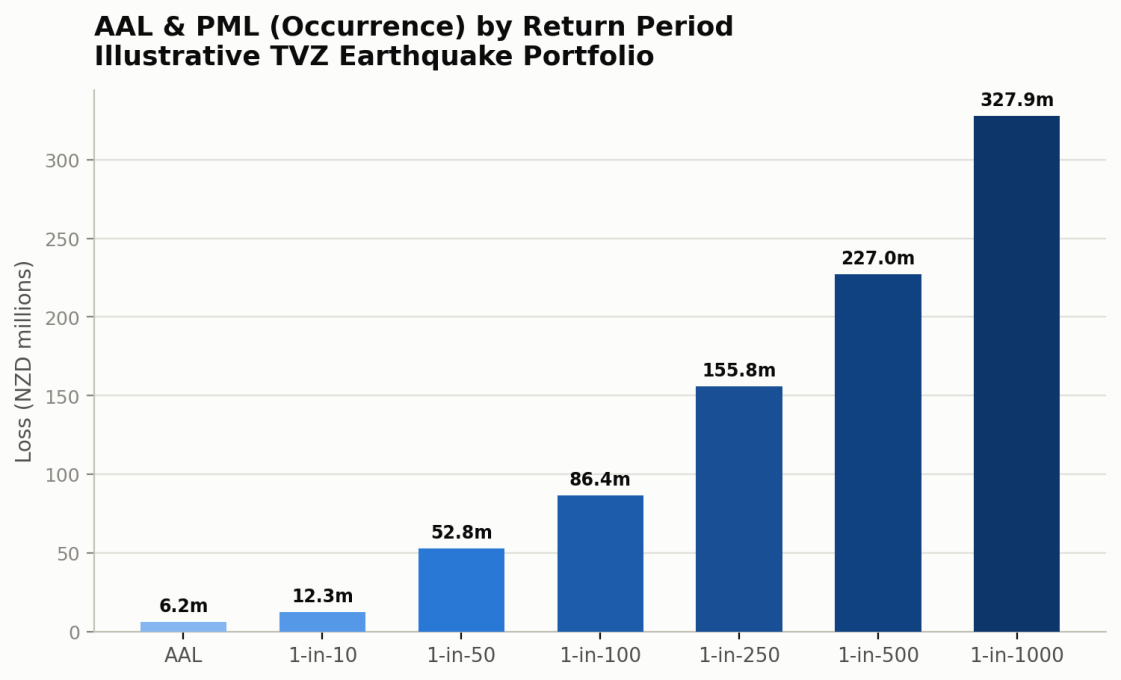


Figure: illustrative AAL alongside PML at increasing return periods for the TVZ earthquake portfolio.

PML (Probable Maximum Loss)

The loss at a specific high return period (e.g. 1-in-100 or 1-in-250 years); used for capital and reinsurance decisions.

- **Example:** reading off the same simulation, the 1-in-250-year PML for the TVZ portfolio is ~**NZD 156m**, and the 1-in-500-year PML is ~**NZD 227m**, these are the numbers an insurer would use to decide how much reinsurance to buy above its own retention, since a single Taupo Rift event at that severity could otherwise wipe out a large share of the book in one year. PML is usually quoted on an occurrence (OEP) basis, as below, though an aggregate (AEP) view is often shown alongside it:

Return period	PML — occurrence (OEP)	PML — aggregate (AEP)
1-in-10	NZD 12.3m	NZD 14.0m
1-in-50	NZD 52.8m	NZD 57.2m
1-in-100	NZD 86.4m	NZD 91.5m
1-in-250	NZD 155.8m	NZD 158.6m
1-in-500	NZD 227.0m	NZD 230.6m
1-in-1000	NZD 327.9m	NZD 330.5m

OEP (Occurrence Exceedance Probability curve)

An EP curve based on the single largest loss event in a year.

- **Example:** if you only care about "what's the worst single earthquake we could be hit by" rather than the whole year's activity, the OEP curve answers that, in my simulation, the largest single modelled event at the 1-in-250-year point costs ~**NZD 156m**, rising to ~**NZD 227m** at 1-in-500. This is the classic basis for PML.

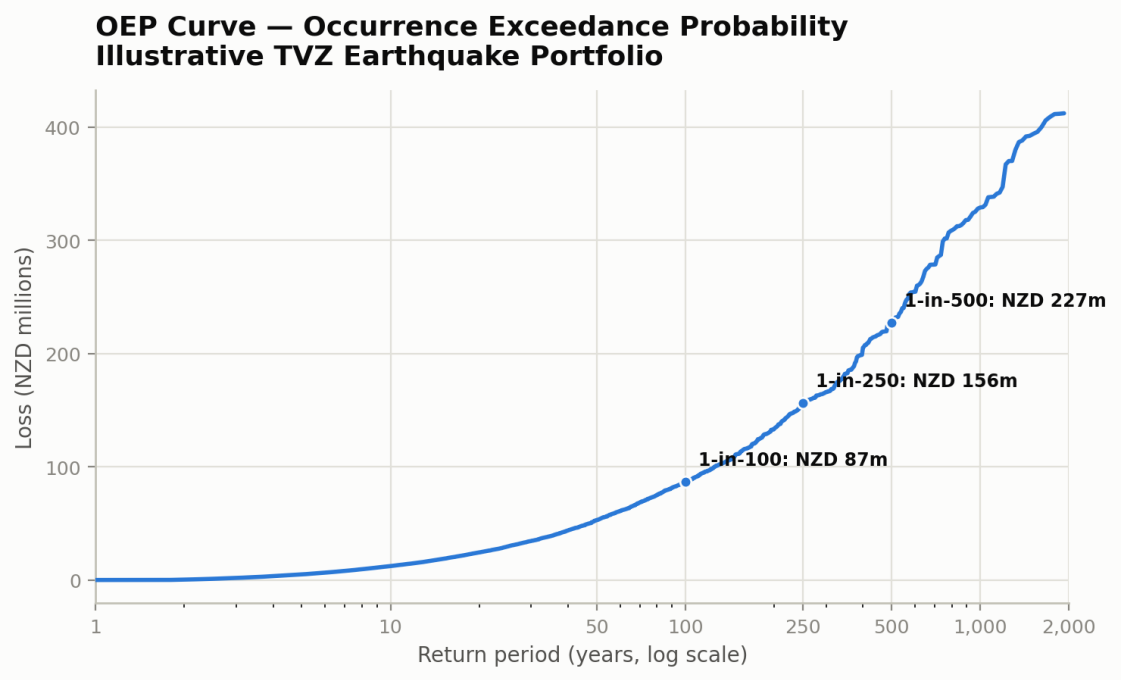


Figure: illustrative OEP curve, loss from the single largest event in a year, by return period.

AEP (Aggregate Exceedance Probability curve)

An EP curve based on the sum of all losses across a year, not just the single largest event.

- **Example:** in a busy TVZ year with two or three moderate quakes rather than one large one, AEP captures the combined toll that OEP alone would miss, in my simulation the 1-in-250-year *aggregate* annual loss is ~**NZD 159m**, very close to the OEP figure here, but the gap between the two curves widens in years with multiple simultaneous or clustered events, which is exactly the scenario AEP is built to capture.

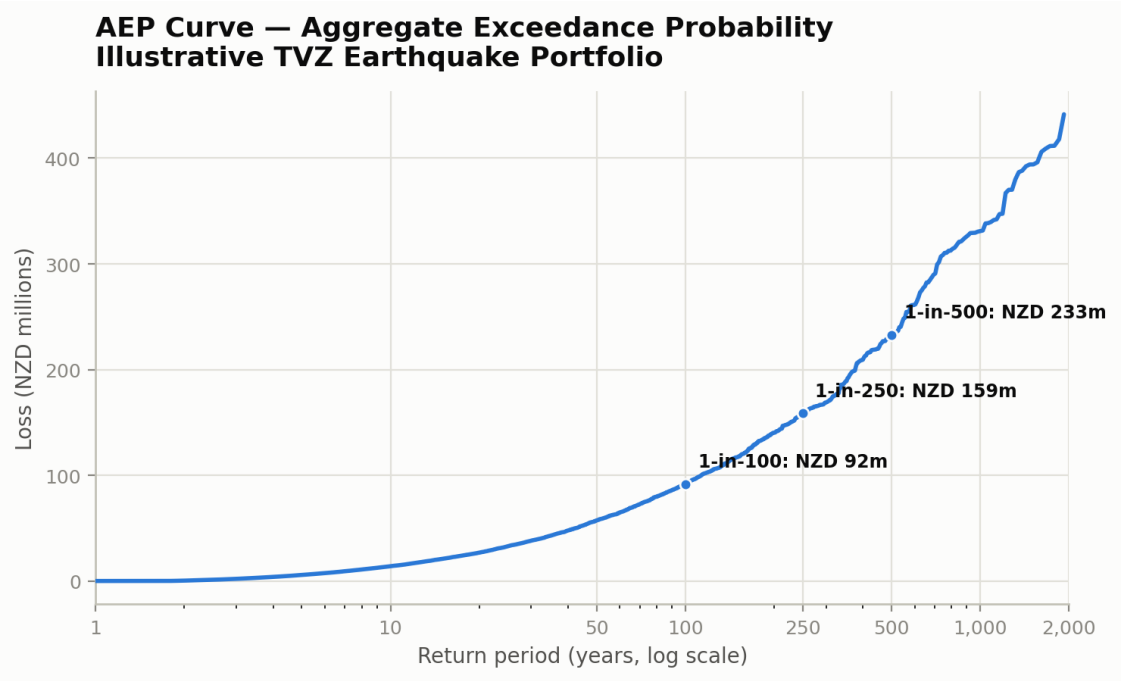


Figure: illustrative AEP curve, total annual loss across all events in a year, by return period.

ELT (Event Loss Table)

A model output listing every simulated event together with its associated loss and probability.

- **Example:** a full ELT for this portfolio would list every event in the stochastic catalogue (potentially tens of thousands of rows); the excerpt below shows a handful of representative TVZ events, from frequent, low-severity ruptures on the Taupo Fault Belt through to the rare, near-total-loss scenario of a full Taupo Rift rupture.

Event ID	Location (TVZ)	Mw	Annual rate	Return period (yrs)	Mean modelled loss
TVZ-00142	Taupo Fault Belt	5.6	0.180	6	NZD 3.2m
TVZ-00098	Reporoa Basin	5.8	0.140	7	NZD 5.6m
TVZ-00211	Waiotapu–Broadlands geothermal	5.9	0.110	9	NZD 7.1m
TVZ-00340	Wairakei–Tauhara field	6.0	0.085	12	NZD 11.4m
TVZ-00087	Rotorua Caldera margin	6.1	0.065	15	NZD 15.8m
TVZ-00265	Kawerau–Edgecumbe Fault	6.3	0.042	24	NZD 26.5m
TVZ-00019	Ngatamariki sector	6.4	0.030	33	NZD 34.9m
TVZ-00402	Tarawera Fault	6.6	0.019	53	NZD 52.3m
TVZ-00155	Taupo Fault Belt (northern)	6.8	0.011	91	NZD 78.6m
TVZ-00311	Matata–Whakatane sector	7.0	0.006	167	NZD 115.0m
TVZ-00450	Central TVZ rift, multi-fault	7.3	0.0028	357	NZD 187.0m
TVZ-00478	Taupo Rift, full rupture	7.6	0.0009	1,111	NZD 312.0m

YLT (Year Loss Table)

A model output listing simulated losses aggregated by simulated year, used to derive AEP curves.

- **Example:** a full YLT for this run would have 50,000 rows, one per simulated year (most showing zero loss, since TVZ-damaging events are relatively rare in any single year). The excerpt below spans that range, from quiet years with no loss, through a moderate single-event year, to the worst simulated year in the whole 50,000-year run.

Sim. year	Events in year	Aggregate annual loss	Largest single event
Yr 41,399	4	NZD 844.7m	NZD 550.0m
Yr 27,440	2	NZD 577.0m	NZD 526.6m
Yr 36,253	2	NZD 557.5m	NZD 550.0m
Yr 22,070	2	NZD 554.4m	NZD 550.0m
Yr 46,859	1	NZD 24.1m	NZD 24.1m
Yr 14,784	3	NZD 9.2m	NZD 5.7m
Yr 8,435	1	NZD 3.0m	NZD 3.0m
Yr 19	1	NZD 0.9m	NZD 0.9m
Yr 16,635	1	NZD 0.3m	NZD 0.3m
Yr 37,411	0	NZD 0	NZD 0
Yr 38,987	0	NZD 0	NZD 0